

AUGUST 2026

INSTITUTIONAL TOKENIZATION:

FROM DIGITAL REPRESENTATION TO MARKET ARCHITECTURE

Regulatory, Infrastructure and Market Architecture
Perspectives for 2026



Strategic Report for the U.S. Securities
and Exchange Commission and
Global Financial Institutions



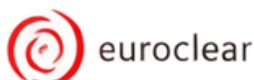
REGULATORS, CENTRAL BANKS & STANDARD SETTERS



REGULATORY AUTHORITIES & SUPERVISORS



GLOBAL FINANCIAL INFRASTRUCTURE & MARKET PARTICIPANTS



ERC-Genesis (ERC-G)

By Monica Elizabeth Pagano

Institutional tokenization is already moving measurable amounts of capital through regulated financial infrastructure. **J.P. Morgan** reports that its **Kinexys Digital Assets** platform has facilitated more than \$1.5 trillion in trading and settlement activity, using tokenized assets including U.S. Treasuries, money-market funds and fixed-income instruments for repo, collateral and margin management.

DTCC, whose depository subsidiary held and serviced \$114 trillion of securities in 2025, successfully processed real production transactions in July 2026 using DTC-tokenized securities with more than 30 financial institutions, covering Treasury/repo DvP, equity DvP, securities lending, collateral pledges, token transfers and CCP margin workflows; DTCC plans to launch the service commercially in October 2026. **BlackRock**, meanwhile, has extended tokenized functionality to institutional money-market funds representing a combined \$311 billion in AUM across 15 markets, while its 2026 Chairman's Letter states that the firm has nearly \$150 billion in AUM connected to digital assets.

In primary markets, **Standard Chartered** issued \$200 million of three-year digitally native floating-rate notes directly through Euroclear's regulated D-FMI in August 2026. These figures are more significant than an aggregate "RWA market cap": they show that tokenization is entering existing institutional pools of securities, cash management, collateral, repo, settlement and regulated issuance, rather than developing solely as a parallel crypto market.

EXECUTIVE SUMMARY

Institutional tokenization is entering a phase distinct from the one that defined its early years. For regulated markets, the relevant question is no longer whether distributed ledger technology (DLT) can technically represent a bond, an equity, a fund interest, a deposit, or a real-world asset.

That capability has been established.

The question in 2026 is an institutional one: under what architecture can such a representation coexist with the instrument's legal identity, the authoritative record of ownership, investor protection, custody, trading, collateral, settlement money, netting, servicing, and operational continuity?

The Financial Stability Board continues to characterize aggregate adoption as limited, albeit growing. However, the pattern emerging in 2026 indicates that the infrastructure capable of supporting institutional volumes is moving from isolated trials into regulated production environments.

The data must be interpreted carefully. The ECB estimated that, at the end of 2025, approximately €2 billion in euro-denominated debt had some form of digital exposure—a figure that remained marginal relative to traditional markets. At the same time, its wholesale settlement trials mobilized approximately €1.6 billion, brought together 64 eligible participants from nine jurisdictions, and covered 58 use cases.

In the United States, on July 15, 2026, DTCC processed live transactions involving tokenized DTC assets with more than 30 institutions, ahead of the planned launch of its service in October. The correct interpretation is not that the tokenized market has already achieved systemic scale, but that systemically important institutions are building the capacity to absorb it.

Institutional investment is concentrating on areas where tokenization can transform a specific economic function: collateral mobility, broader operational access, reduced reconciliation, programmable servicing, distribution, coordinated settlement, and more efficient use of balance sheets and liquidity. The ECB has observed an initial concentration in debt, money market funds, and collateral and repo markets, while more complex instruments remain less developed.

DTCC is building around collateral and post-trade functions; BNY and State Street around transfer agency, custody, and servicing; and Citi has extended the model to private markets. This pattern matters because it shifts the analysis away from the token as a technological object and toward the market function that must remain legally valid and operationally resilient.

The first market-structure conclusion is that tokenization does not necessarily create liquidity. An additional digital representation may fragment fungibility, price discovery, and market depth if it creates a parallel market.

The Nasdaq/DTC model approved by the SEC is significant because it preserves the same security, CUSIP, ticker, rights, order book, and order priority, while moving the choice of format to the post-trade stage.

Under this design, technological innovation neither redefines the instrument nor duplicates its market; it changes how regulated infrastructure represents and processes it.

The second conclusion is that instantaneous settlement does not automatically mean optimal settlement. Atomicity may reduce certain exposures and reconciliation requirements, but continuous gross settlement may increase intraday liquidity needs and eliminate the efficiencies provided by netting.

For this reason, the BIS continues to investigate liquidity-saving mechanisms. Standard Chartered, HSBC, and Swift have demonstrated a hybrid model in which tokenized obligations are coordinated, matched, and netted through new infrastructure before final settlement takes place through existing systems.

The emerging institutional design does not eliminate market functions. Instead, it redistributes which functions must be programmed, which must interoperate, and which should remain within infrastructures that provide finality, netting, and consolidated controls.

This report uses ERC Genesis as an architectural case study, rather than as a premise for the analysis or as a proposal to replace existing infrastructure. The dossier describes a legal and technical architecture designed to link an asset's legal and registry identity to a foundational digital reference and to use that reference as the basis for structuring securities, primary and secondary issuances, collateral, tranching debt, mortgage-backed structures, MBS, ABS/REIT instruments, income distribution, automated payments and guarantees, KYC/AML, custody integration, oracles, and multichain interoperability.

The regulatory significance of this proposal does not lie in the existence of another token standard, but in a broader question: if trading, custody, settlement, and interoperability infrastructure advances faster than the architecture that defines the asset and its associated rights, ***how can a consistent source of truth be preserved throughout the entire lifecycle?***

BUIDL LIVE BlackRock USD Institutional Digital Liquidity Fund ASH NAV/PRICE \$2.64B \$1.00 TOKENIZED TREASURY ETHEREUM BLACKROCK / CONTRACT	BENJI SNAPSHOT Franklin OnChain U.S. Government Money Fund ASH NAV/PRICE \$1.70B \$1.00 TOKENIZED TREASURY STELLAR / POLYGON FRANKLIN TEMPLETON / CONTRACT	JAAA SNAPSHOT Janus Henderson AAA CLO ETF ASH NAV/PRICE \$30.00B \$50.57 CLO / CREDIT NYSE ARCA JANUS HENDERSON / CONTRACT	JLTXX SNAPSHOT JPMorgan USD Liquidity LTVNAV Fund ASH NAV/PRICE \$792.0M \$1.00 TOKENIZED RMF ONYX DIGITAL ASSETS JPMORGAN ONYX / CONTRACT
OUSG SNAPSHOT Ondo Finance US Treasury ASH NAV/PRICE \$850.0M \$109.50 TOKENIZED TREASURY ETHEREUM / SOLANA ONDO FINANCE / CONTRACT	GS-DAP SNAPSHOT Goldman Sachs Digital Asset Platform Bond ASH NAV/PRICE \$500.0M \$100.00 DIGITAL BOND GS DAP (PRIVATE) GOLDMAN SACHS / CONTRACT	USDY SNAPSHOT Ondo U.S. Dollar Yield Token ASH NAV/PRICE \$420.0M \$1.08 YIELD-BEARING STABLECOIN ONDO FINANCE / CONTRACT	SCRT-10X SNAPSHOT Securitize Markets Digital Securities ASH \$1.20B DIGITAL SECURITIES PLATFORM MULTI-CHAIN SECURITIZE / CONTRACT

Functional comparison is essential. Swift coordinates monetary obligations; Chainlink provides data, compliance, and interoperability; DTCC represents and processes assets held in custody at DTC; Canton provides an institutional network with privacy features; and custodians and transfer agents perform control, recordkeeping, and servicing functions. Genesis seeks to operate at a different layer: asset definition and anchoring, instrument structuring, and lifecycle rules that must subsequently be integrated with those infrastructures.

That distinction must be subject to legal, technical, and regulatory validation: which record prevails, what rights are created, who is accountable for each function, how different states are reconciled, and which components actually exist as deployable software.

Developments in 2026 suggest that the next stage of institutional tokenization will not be determined by the capacity to issue more digital representations. It will be determined by the quality of the architecture connecting instruments, rights, records, intermediaries, infrastructure, and settlement without weakening investor protection or market integrity.

For regulators, the priority is to identify where a regulated function changes and who remains accountable for it.

For institutions, the priority is to avoid architectures that automate transfers before determining what is being transferred, which record is authoritative, and how rights will be maintained throughout the entire lifecycle. ERC Genesis is assessed in this document exclusively within the context of this architectural challenge.

The screenshot displays the ERC-Genesis website. The top navigation bar includes links for Home, Market RWA, Compliance Map, Framework, Research, IP & Regulatory, About, and an Inquiry button. The main content area is divided into a sidebar for regions and a main panel for the selected region.

Region Selection:

- 01 North America (Selected)
- 02 Latin America
- 03 Europe
- 04 Asia-Pacific
- 05 Africa
- 06 International Bodies

United States Content:

US United States 4 SOURCES

Dual regulatory framework (SEC + CFTC). Crypto assets may qualify as securities under the Howey Test. The SEC Crypto Task Force (2025-2026) is developing guidance on tokenized real-world assets. No comprehensive federal crypto law enacted as of 2026.

Category	Source	Action
REGULATOR	SEC — Crypto Assets (Official Topic Page)	↗
FRAMEWORK	SEC — Crypto Task Force (Official Page)	↗
REGULATOR	CFTC — Digital Assets (Official Page)	↗
GUIDANCE	OCC — Interpretive Letter 1179: Crypto Custody & Stablecoins (2021)	↗
AML/KYC	FinCEN — Application of FinCEN Regulations to Certain Business Models (2019)	↗
GUIDANCE	Federal Reserve — Joint Statement on Crypto-Asset Risks (2023)	↗

WALL STREET BLOCKCHAIN

U.S. Treasury | Offered by Securitize Markets

VanEck Treasury Fund

Earn stable yield with daily liquidity by investing in short-term U.S. Treasuries via blockchain.

Asset Class	Fixed Income
Min. Investment	\$100k
Investor Type	Qualified Purchaser
Redemptions	Instant

U.S. Treasury | Securitize Markets Exclusive

BlackRock USD Institutional Digital Liquidity Fund (BUIDL)

The first tokenized fund issued on a public blockchain by the world's largest asset manager.

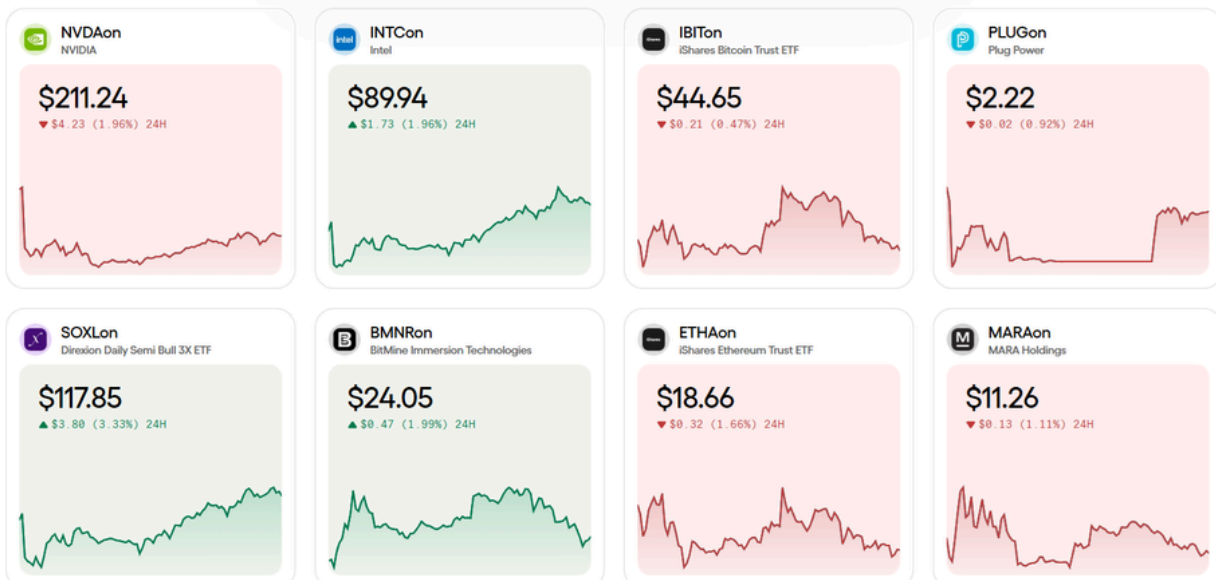
Asset Class	U.S. Treasury
Min. Investment	\$5M
Investor Type	Qualified Purchaser
Redemptions	Instant

Private Equity | Offered by Securitize Capital

Hamilton Lane Senior Credit Opportunities Securitize Fund

Evergreen private credit fund seeking risk-adjusted returns and attractive floating cash yield.

Asset Class	Senior Credit
Min. Investment	\$10K
Investor Type	Qualified Purchaser
Redemptions	On-Demand



SOEZ

Franklin Solana ETF

[Learn More](#)

Asset Class	Alternatives
Inception Date	12/03/2025
Dividend Frequency	Monthly
NAV As of 08/24/2026	\$16.59

XRPZ

Franklin XRP ETF

[Learn More](#)

Asset Class	Alternatives
Inception Date	11/24/2025
Dividend Frequency	No
NAV As of 08/24/2026	\$16.03

EZPZ

Franklin Crypto Index ETF

[Learn More](#)

Asset Class	Alternatives
Inception Date	02/20/2025
Dividend Frequency	No
NAV As of 08/24/2026	\$19.91

EZBC

Franklin Bitcoin ETF

[Learn More](#)

Asset Class	Alternatives
Inception Date	01/11/2024
Dividend Frequency	No
NAV As of 08/24/2026	\$45.55

EZET

Franklin Ethereum ETF

[Learn More](#)

Asset Class	Alternatives
Inception Date	07/23/2024
Dividend Frequency	No
NAV As of 08/24/2026	\$18.71





Self-Regulatory Organizations; NYSE Arca, Inc.;
Notice of Filing and Immediate Effectiveness of
Proposed Rule Change to Amend the Exchange's
Rules to Enable the Trading of Securities on the
Exchange in Tokenized Form

National Securities Exchanges, NYSE Arca (NYSEARCA)

NYSE Arca already has a framework for trading tokenized securities and ETFs through DTC. However, there is not yet a publicly available list of "NYSE-listed tokenized funds" that retail investors can purchase directly on-chain.





At present, Nasdaq does not provide a public catalog of tokenized securities, meaning there is no public interface where investors can view listings such as “tokenized AAPL” or “tokenized MSFT.” Nasdaq has not yet issued the first Equity Trader Alert identifying the eligible tickers, although it has stated that updated lists will be published periodically through these alerts.

Tokenized positions are created after trade execution and delivered to DTC-registered wallets. Consequently, retail investors cannot currently purchase Nasdaq tokenized securities directly through a wallet. Access to the tokenized format requires either eligibility to participate in DTC’s tokenization program or the use of an intermediary participating in the program.

Market Vol	Top Matched	Top Matched NTX	Top Matched PSX
U.S. Market Volume			
08/25/2026			
Volume	Trades	Dollar Volume	Set Default Tab
Equities Market Volume			
Equities	Tape A	Tape B	Tape C
Nasdaq	27,827,477	25,841,417	167,995,483
Nasdaq %	10.5%	9.1%	12.0%
Nasdaq Texas	611,521	926,869	1,841,898
Nasdaq Texas %	0.2%	0.3%	0.1%
PSX	358,191	262,627	676,904
PSX %	0.1%	0.1%	0.0%
FINRA/TRF	Tape A	Tape B	Tape C
FINRA/Nasdaq TRF™ Carteret	103,528,898	100,698,046	620,742,683
FINRA/Nasdaq TRF™ Carteret %	39.0%	35.5%	44.5%
FINRA/Nasdaq TRF™ Chicago	1,124,567	1,353,658	4,542,534
FINRA/Nasdaq TRF™ Chicago %	0.4%	0.5%	0.3%
Total	265,537,716	283,867,253	1,395,304,611

Data is updated every 1 minute. Last Update as of 08/25/2026 09:42

The Nasdaq Stock Market LLC; Notice of Filing of a Proposed Rule Change, as Modified by Amendment No. 2, to Amend the Exchange’s Rules to Enable the Trading of Securities on the Exchange in Tokenized Form

From Experiments to Market Infrastructure

The institutional thesis has changed. In June 2025, the BIS summarized its next-generation vision around a “trilogy” of tokenized central bank reserves, tokenized commercial bank money, and tokenized sovereign bonds. Project Agorá, published

in May 2026, technically demonstrated that tokenized commercial bank deposits and tokenized central bank reserves can be combined on a programmable platform for wholesale, multicurrency, and potentially continuous settlement. This brings tokenization much closer to the architecture of money and capital markets than to the original world of crypto trading.

At the same time, the International Monetary Fund identifies a broader economic rationale: tokenization may reduce friction in issuance, servicing, redemption, and trading; enable more flexible settlement; and reduce certain counterparty and search frictions. However, it may also increase interconnectedness, infrastructure concentration, and risks arising from code and programmable contracts. Institutional investment is therefore justified by lifecycle reengineering, not by blockchain per se.

The best indicator of demand is not currently the total amount of announced RWAs, but where the institutions responsible for market infrastructure are deploying capital and operational capacity.

DTCC is bringing tokenization to a depository that holds more than \$114 trillion in assets. Its July trial included collateral pledges, securities lending, Treasury/repo DvP, equity DvP, delivery-versus-delivery, token transfers, and CCP margin workflows.

This demonstrates that the institutional priority is to use the new form of representation to improve financing, collateral, settlement, and post-trade functions, rather than simply issuing financial NFTs.

Banking demand reflects the same pattern. J.P. Morgan tested a cross-chain DvP transaction with Chainlink and Ondo in which the asset—tokenized U.S. Treasuries—was held on a public network, while the cash leg consisted of USD deposits within Kinexys. JPM Coin is now an institutional deposit token designed for 24/7 payments, collateral, and settlement. BNY began by representing deposits on-chain for collateral and margin workflows, while State Street is building a platform that combines wallets, custody, cash, and tokenized products.

Custody and transfer agency are becoming digital infrastructure, not disappearing. In July 2026, BNY launched its Digital Transfer Agency to support traditional and digitally native funds jointly, across multiple blockchains and jurisdictions, with a fund's legal books and records capable of residing on a public blockchain. State Street intends to provide issuance, administration, and custody services for tokenized funds from Luxembourg within the same operating model used for traditional funds. Citi, meanwhile, combines the roles of issuer and custodian in its Digital Depositary Receipts representing private-company shares through SIX's regulated infrastructure.

The result is a new institutional stack:

Asset/legal rights → issuance and structuring → authoritative record → custody and servicing → trading and liquidity → collateral and financing → cash leg → settlement and netting → cross-chain interoperability, data, and compliance.

Today, none of the major platforms controls this entire stack. This fragmentation of functions is both the interoperability problem and the commercial opportunity. The SEC explicitly distinguishes between securities tokenized by the issuer itself and representations created by third parties, precisely because the technology can represent legally distinct rights. DTCC retains its official books and records; BNY connects the digital record to transfer agency; and Swift connects deposits issued on different bank ledgers. The institutional challenge is to ensure that all these states remain synchronized.

Regulation: The Perimeter Is Tightening

The most important regulatory signal of 2026 is not that securities laws have yielded to technology. It is the opposite: technology is being incorporated into regulated markets on the premise that economic substance, investor rights, record integrity, and intermediary responsibilities remain decisive. This shifts the center of gravity of the analysis.

The regulatory question is no longer “Is it on a blockchain?” but rather “What right does it represent, who issues it, which record evidences ownership, who holds it in custody, where is it traded, and how are its obligations enforced?”

In the United States, the joint Staff Statement issued on January 28, 2026, by the SEC’s Divisions of Corporation Finance, Investment Management, and Trading and Markets defines a tokenized security as an instrument that already falls within the federal definition of a security and whose ownership is maintained, in whole or in part, through one or more crypto networks.

The taxonomy distinguishes between issuer-sponsored and third-party-sponsored models because technically similar structures may confer different legal rights. For institutional architecture, the consequence is direct: the digital representation cannot be analyzed separately from the master securityholder file, the issuer, the transfer agent, the holder’s rights, and—in third-party structures—counterparty or insolvency risk.

The SEC's March 18, 2026 approval of Nasdaq's proposal adds a second component. The model permits securities to be traded in tokenized form while preserving their economic identity and rights—including the same CUSIP, ticker, and access to the same order book—and uses DTC for post-trade processing. Its significance is not simply that the SEC permitted “tokenized trading.”

The significance is that the design avoids creating a second, economically equivalent security and preserves the existing infrastructure for price discovery, investor protection, and post-trade processing. It is a precedent for integration, not replacement.



U.S. Securities and
Exchange Commission

INTERPRETIVE RELEASE

Application of the Federal Securities Laws to Certain Types of Crypto Assets and Certain Transactions Involving Crypto Assets

Public Comments

[SUBMIT A COMMENT ON S7-2026-09](#)

[VIEW RECEIVED COMMENTS](#)

On March 17, the SEC and CFTC also issued a Commission-level interpretation addressing the treatment of different categories of crypto assets, effective March 23. This is important in establishing a more coherent perimeter for digital markets, but it should be conceptually distinguished from the Statement on tokenized securities: a tokenized security remains a security because of its economic and legal nature. It does not become a different regulatory category merely because it uses DLT.

In derivatives markets, the CFTC opened a commercially significant area in December 2025: tokenized collateral. Its guidance covers tokenized RWAs such as U.S. Treasuries and money market funds and requires an asset-by-asset assessment of legal enforceability, segregation, custody and control, haircuts, valuation, and operational risk.

This supports the use of tokenized collateral, while also confirming that a blockchain-based representation has institutional value only when there is certainty regarding the underlying asset, control, valuation, and legal enforceability.



Release Number 9146-25

Acting Chairman Pham Announces Launch of Digital Assets Pilot Program for Tokenized Collateral in Derivatives Markets

CFTC Issues New Guidance to Provide Regulatory Clarity, Eliminates Outdated Requirements that Hurt Innovation

December 08, 2025

WASHINGTON — Commodity Futures Trading Commission Acting Chairman Caroline D. Pham today announced the launch of a **digital assets pilot program** for certain digital assets, including BTC, ETH, and USDC, to be used as collateral in derivatives markets; **guidance on tokenized collateral**; and **withdrawal of outdated requirements** given the enactment of the GENIUS Act. Today's announcement marks a significant milestone in the expanded adoption of digital assets in regulated markets with appropriate guardrails, and follows the tokenized collateral **initiative** Acting Chairman Pham launched in September as a part of the CFTC's **Crypto Sprint** to implement recommendations in the President's Working Group on Digital Asset Markets **report**.

Europe is following a different but convergent path. MiCA, fully applicable since December 2024, does not regulate tokenized securities or tokenized deposits as though they were generic crypto-assets; those instruments remain subject to banking and securities law.

The DLT Pilot Regime permits experimentation involving shares, bonds, and UCITS under specific exemptions from MiFID II and CSDR. In June 2025, ESMA proposed making the regime permanent and more flexible after observing limited initial uptake but growing interest.





EUROPEAN CENTRAL BANK | EUROSISTEM



27 January 2026

Eurosistem to accept marketable assets issued in central securities depositories (CSDs) using distributed ledger technology (DLT) as eligible collateral for Eurosistem credit operations as of 30 March 2026

Further work is exploring ways to expand eligibility to assets issued and settled entirely on DLT networks

Decision reflects Eurosistem's commitment to innovation and fosters technological progress in European financial markets



The FCA and Bank of England (The Bank) are asking for views on how to support the safe adoption of tokenisation in UK wholesale financial markets.



FCA and Bank of England set out shared vision for tokenisation in UK wholesale markets

Europe's 2026 leap is monetary and infrastructural. Since March 30, the Eurosystem has accepted certain marketable assets issued through DLT services operated by CSDs as collateral, provided they satisfy the existing eligibility criteria. Pontes, whose pilot phase is scheduled for the third quarter of 2026, will connect DLT platforms to TARGET Services for settlement in central bank money.

Appia will develop, through 2028, a blueprint for an integrated European tokenized financial market.

In the United Kingdom, the FCA and the Bank of England published a joint vision for wholesale tokenization on May 18, 2026, focusing on tokenized securities, prudential treatment, collateral, and settlement instruments. Sixteen firms were working with the authorities within the Digital Securities Sandbox on live issuance and settlement.

The Bank of England also announced a future synchronization service, targeted for 2028, alongside work to enable tokenized equivalents to be accepted as collateral by CCPs and in central bank operations.

Institutional Tokenization Regulatory Timeline

Key regulatory developments and implications for market infrastructure

Jurisdiction / action	Date	What changes	Institutional implication
SEC — Statement on Tokenized Securities	28/1/2026	Distinguishes issuer-sponsored, custodial third-party, and synthetic models; the format does not alter the application of securities laws.	Ownership architecture and the master record become first-order considerations.
SEC/CFTC — Joint interpretation	17/3/2026	Establishes a taxonomy of crypto assets and coordinates SEC/CFTC treatment.	Reduces uncertainty around the crypto regulatory perimeter; it does not replace the laws applicable to tokenized securities.
SEC — Nasdaq tokenized trading	18/3/2026	Permits a tokenized format while preserving the same CUSIP, ticker, rights, and order book under the DTC Pilot.	Provides the first clear pathway toward tokenization without fragmenting national-market liquidity.
CFTC — Tokenized collateral guidance	8/12/2025	Allows tokenized Treasuries and money market funds to be assessed as collateral and sets criteria for enforceability, custody, valuation, and risk.	Collateral emerges as a use case with immediate institutional ROI.
EU — DLT Pilot / ESMA review	25/6/2025	ESMA recommends making the regime permanent and more flexible.	The EU is moving from a temporary sandbox toward a more stable regulatory infrastructure.

Eurosystem — DLT collateral	30/3/2026	Certain securities issued using DLT through CSDs become eligible as collateral.	Central-bank eligibility increases economic utility and potential liquidity.
Eurosystem — Appia/Pontes	11/03/2026 / Q3 2026	Settlement of DLT assets in central-bank money and a roadmap for an integrated market.	Addresses one of the principal bottlenecks: the institutional cash leg.
UK — FCA/BoE wholesale vision	18/5/2026	Covers the DSS, tokenized collateral, settlement, and a future synchronization service.	Provides an explicit signal to move from pilots to supervised production.

Comparative regulatory assessment. The United States, the European Union, and the United Kingdom are not converging on a single technological network, but they are converging on a common set of invariants: economic substance, legal rights, custody and control, market integrity, settlement finality, operational resilience, and accountability remain unchanged despite a shift in technological rails.

Accordingly, the principal design risk is not choosing the wrong blockchain; it is introducing a digital representation whose relationship with the underlying instrument, the authoritative record, or regulated intermediaries is ambiguous.

Market Architecture: Infrastructure, Standards, and Risks

Primary issuance. Issuance can become a significantly more programmable chain of processes: instrument creation, allocation, KYC, subscription, payment, issuance, transfer restrictions, and servicing can all be coordinated through smart contracts. Standard Chartered recently issued \$200 million in three-year floating-rate notes directly on Euroclear's D-FMI, connecting DLT with existing regulated issuance, settlement, and servicing processes. Citi is pursuing a similar approach in private markets through Digital Depositary Receipts representing private-company shares.

Secondary markets. The challenge is not making the token transferable, but preserving price discovery and liquidity concentration. Nasdaq and DTC offer one response: the same order book and the same security, but a different settlement format. DTC has also validated conversions between traditional and tokenized representations and is pursuing a multichain strategy. This will likely become one of the dominant architectures for liquid public assets:

tokenization as an extension of existing infrastructure, not as a parallel market.

Custody and asset servicing. In institutional markets, private keys do not, by themselves, replace the functions performed by custodians. Legal control, segregation, key recovery, corporate actions, official books and records, reconciliation, reporting, and insolvency scenarios must all be addressed. The SEC expressly distinguishes between direct and indirect rights and identifies bankruptcy risks in third-party structures; the CFTC requires an assessment of custody and control over collateral; and BNY and State Street are integrating these elements into their traditional servicing models.

Settlement and netting. Tokenization enables programmable DvP/PvP and near-instantaneous settlement, as demonstrated by Kinexys/Chainlink and Project Agorá. However, it does not eliminate the economics of netting. The BIS notes that RTGS systems have intraday liquidity requirements that justify liquidity-saving mechanisms.

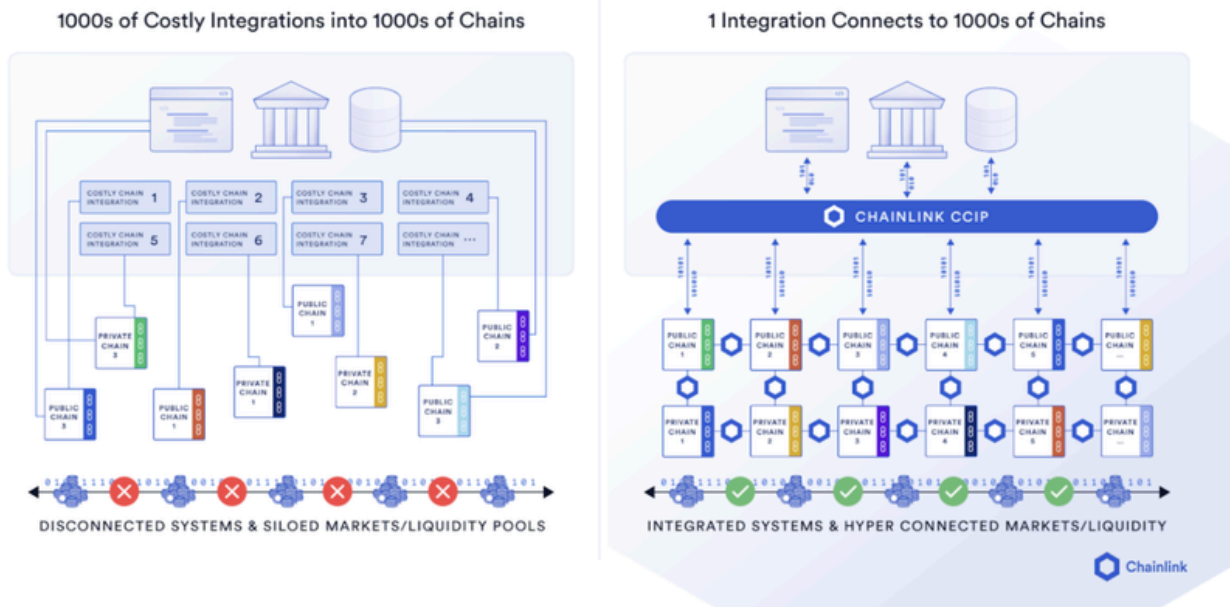
Significantly, Swift chose to match and net tokenized obligations between Standard Chartered and HSBC before final settlement.

The future will not necessarily be “everything at T+0”; it will involve settlement optimized according to risk, liquidity, and balance-sheet considerations.

Interoperability. This is the most visible bottleneck. DTC is developing a multichain strategy; Chainlink CCIP connects more than 70 networks; and CRE is being used on DTCC’s Collateral AppChain for data, valuation, eligibility, margining, and settlement workflows. Canton is pursuing interoperability with institutional-grade privacy, while the ECB is explicitly considering whether Europe should ultimately operate through a shared ledger or multiple interconnected networks.

This is the most visible bottleneck. DTC is developing a multichain strategy; Chainlink CCIP connects more than 70 networks

Efficiently Connecting Banks to Blockchains via Existing Banking Standards



Technical interoperability, however, is not equivalent to legal interoperability of the instrument.

A message may pass between Chain A and Chain B without certainty that both systems recognize the same legal right. For a regulator or a financial market infrastructure, the relevant questions are different:

Is it the same security or a different entitlement?

Which record is legally constitutive?

What happens if on-chain and off-chain records diverge?

How are corporate actions processed?

How can a representation be revoked or updated?

Who bears the loss if the technical layer and the legal record become unsynchronized?

The SEC's taxonomy demonstrates precisely that similar digital representations can produce different legal consequences.

For that reason, a token standard should not be confused with a market standard.

ERC-3643 can encode permissioning, and a layer such as Chainlink ACE can enforce identity, KYC/AML, and jurisdictional rules. However, a regulated market also requires authoritative records, governance, data standards, custody arrangements, settlement conventions, corporate-action processing, exception management, and reconciliation rules.

Useful standardization is not the standardization that merely makes token formats uniform; it is the standardization that enables different institutions to interpret and execute the same rights and obligations consistently.

In cloud infrastructure, the role is different. **The cloud is an enabling layer, not necessarily the authority governing the security.** Azure Confidential Ledger, for example, provides a permissioned, append-only ledger with cryptographic integrity proofs, making it useful for audit trails and sensitive data.

Microsoft also participated in DTCC's production transactions in July.

Nevertheless, ***the asset's legal validity continues to depend on the issuer, transfer agent, registry, CSD, custodian, and applicable law—not on the cloud provider.***

The risks that should appear in the risk register of any institutional program are therefore broader than smart-contract risk. The FSB identifies liquidity and maturity mismatch, leverage, asset quality and pricing, interconnectedness, and operational fragility.

Additional risks include legal identity, enforceability, custody and control, AML/KYC, oracle and data risk, privacy, cybersecurity, key management, blockchain outages and forks, bridge and interoperability risk, and potential concentration among a small number of infrastructures with strong network effects.

ERC Genesis as a Case Study in Instrument Architecture

The evaluation of ERC Genesis requires the proposal to be classified correctly. Genesis does not describe merely a token or an issuance platform.

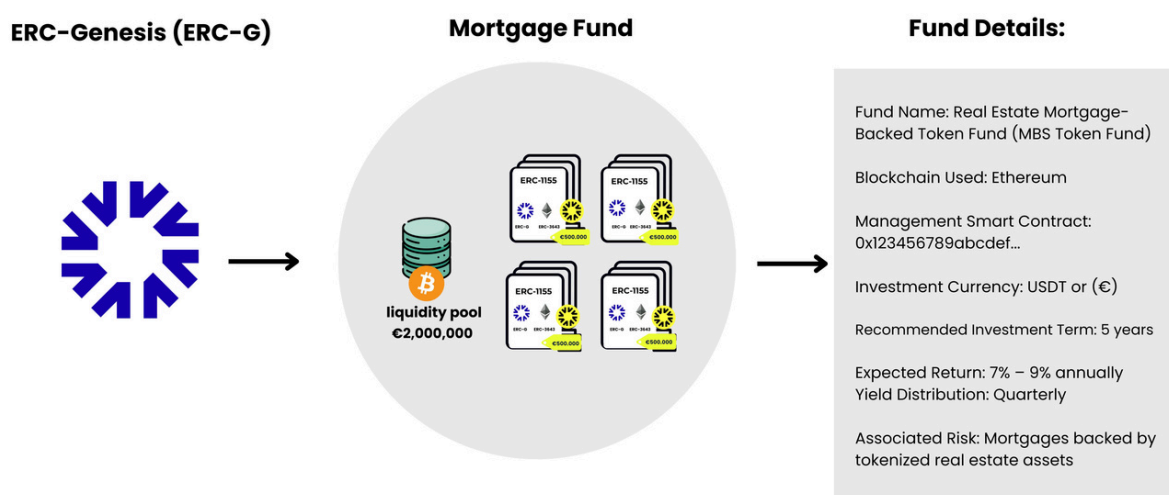
It describes a legal and technical lifecycle architecture for physical, registry-based, and financial assets.

Its central mechanism is a foundational Genesis Token, described as an ERC-1155 token with a supply of one, cryptographically linked to the asset's legal identity through registry information, hashes, and certified documentation. Regulated tokens may be issued against this reference under ERC-3643, subject to KYC/AML requirements, jurisdictional restrictions, and traceability controls.

For a regulatory audience, the critical question is not which ERC standard is used, but whether this mechanism can maintain a legally reliable relationship among the asset, the instrument, the authoritative record, and the digital representation.

The design also contemplates bidirectional interoperability with public and notarial registries, custody integration, collateralization rules, governance logic, automated payment flows, and multichain interoperability.

The protocol establishes that derived representations must retain a verifiable reference to the Genesis Token and links their status within the system to the continued existence and registry status of that foundational reference.



Conceptually, this seeks to address a problem made more visible by developments in 2026: how to prevent the instrument's programmability from becoming detached from the legal, registry, and economic reality it is intended to represent.

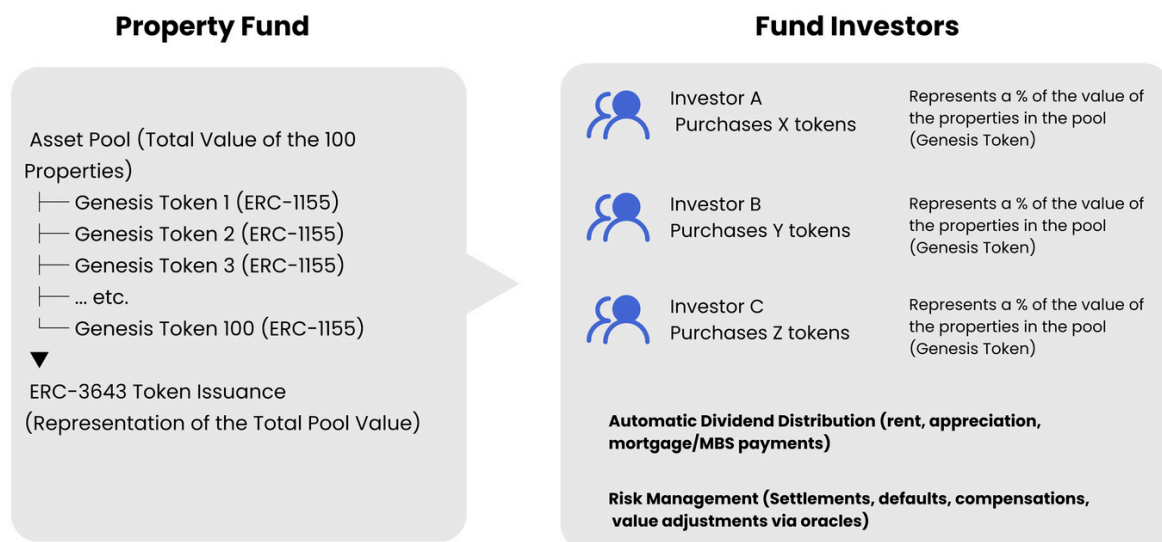
From a capital-markets perspective, the stated scope extends beyond initial tokenization.

The dossier includes procedures for securities issuance, primary and secondary markets, mortgage-backed funds, MBS, tranching debt, automated income distribution, automatic price adjustment, ABS, REIT structures, fund collateral, guarantees, and oracles. It also includes a procedure adapted to the United States and other jurisdictions for using a mortgage or deed of trust as collateral for a tokenized issuance.

These capabilities are relevant because structured finance requires rights, priorities, collateral, waterfalls, and restrictions to remain consistent throughout the entire lifecycle. They are also precisely the areas in which regulatory and operational review must be most rigorous.

The contribution of Genesis, therefore, should not be framed merely as “defining what moves.”

The problem is broader: defining what legally exists, which rights and obligations it creates, how it may be structured or tranching, what collateral supports it, which events alter its status, how its cash flows are distributed, and which representation must be regarded as valid at each stage.



The Genesis Token links the physical asset, the real right, and the representative token.

This is a question of instrument architecture before it is a question of blockchain technology.

Its fit within the market must be assessed cautiously.

The ECB observes that DLT adoption is currently concentrated in straightforward debt instruments, money market funds, and collateral, while equities and complex instruments show lower penetration.

Genesis devotes a significant part of its design to registry-based assets and structured finance.

The greater the instrument's complexity, the greater the need to demonstrate enforceability, bankruptcy treatment, authoritative recordkeeping, custody and control, disclosure, valuation, servicing, and operational resilience.

Functional Comparison: The Problem Addressed by Each Layer
Comparative matrix of functions, scope, and institutional maturity

Capability	ERC Genesis	DTC / FMI tokenization	Chainlink / interoperability	Tokenized deposits / Swift	Canton / institutional network	Custodian / TA platforms
Foundational asset identity linked to registries/documentation	An explicit core element of the design.	DTC maintains the security or entitlement and its official books and records for assets already held in custody.	Can synchronize data and records, but does not seek to act as the issuer or legal registry for each asset.	Not its primary focus; it coordinates monetary obligations.	Provides network infrastructure, privacy, and interoperability.	Can serve as the legal source of ownership for specific products, as in BNY Digital TA.
Issuance of tokenized securities	Yes, including primary and secondary issuance.	Yes, for eligible DTC assets within its service.	Provides enabling infrastructure, but acting as issuer is not its primary function.	Not the primary objective.	The network can support issuance; implementation depends on applications and issuers.	Yes, in specific fund and securities models.
Tranches / MBS / ABS / structured finance	Explicitly contemplated by the design.	Potentially supported in the future, although the public pilot initially focuses on liquid assets.	Can provide data, oracles, and workflow capabilities; structuring is not its primary focus.	No. Its initial focus is commercial bank money.	Can host financing and repo applications, but does not itself define the instrument.	Some platforms support structured products, depending on the specific product and servicer.
Collateral / programmable guarantees	Yes.	A strategic focus for DTCC.	Strong data and orchestration capabilities.	Can provide the cash leg.	Strong use case in repo and collateral mobility.	Custody, control, and valuation are core functions.

KYC/AML and transfer restrictions	Embedded in the method.	Implemented through DTC's regulated participant framework.	ACE encodes KYC/AML and cross-chain identity requirements.	Preserves existing banking controls.	Supports private or permissioned architecture depending on the application.	An inherent part of the regulated custody and servicing model.
Multichain / oracles	Both are contemplated.	DTC is pursuing a multichain strategy.	Core capability: CCIP, CRE, and data/oracle services.	Swift provides interoperability between bank ledgers; it is not an oracle layer.	Interoperability and privacy are core features of the network.	They typically integrate third-party networks.
Settlement / proprietary liquidity	Designs workflows, but requires integration with FMI, exchanges, and money rails.	A core strength.	Orchestrates DvP; it does not necessarily replace the cash leg or the CSD.	A core strength in the money leg.	Supports financing and settlement applications.	Provides custody and servicing; liquidity depends on the venue and product.
Public maturity in institutional production	Must be demonstrated through due diligence and a pilot; the reviewed dossier describes the methodology, architecture, and proposed implementation.	High and increasing: live production activity in July 2026.	Institutional integration already used in trials and within DTCC AppChain.	Live production involving Standard Chartered and HSBC.	Active institutional use cases and DTC integration.	BNY and Citi already offer live services; State Street is progressing toward production.

The comparison with DTCC, Swift, Chainlink, Canton, and custodians must avoid creating a false equivalence. These infrastructures perform different—and, in many cases, regulated—functions. Genesis does not need to replace them for its architectural hypothesis to be relevant.

Subject to validation, its most plausible position is upstream: asset origination, legal and registry anchoring, instrument structuring, and programmable lifecycle management, followed by integration with trading venues, custodians, financial market infrastructures, payment rails, and interoperability networks.

This is an analytical inference based on the functional division observed in the market and the scope described in the dossier.

The Standard Chartered/HSBC/Swift case helps define that boundary. On August 19, 2026, the two banks conducted the first live interbank transfer of tokenized deposits over Swift's ledger. The new infrastructure coordinated the obligations, matching, and netting, while final settlement remained within existing systems.

That case addresses the cash and payment layer.

It does not seek to define the legal structure of an MBS, the tranches of an issuance, the relationship between a mortgage and its registry record, the collateral supporting a security, or the waterfall of an ABS.

This distinction matters because a complete institutional architecture must coordinate both layers without conflating them.

Standard Chartered's issuance of \$200 million in digitally native notes on Euroclear's D-FMI the following day illustrates the same phenomenon from the securities side: digital-money interoperability and digital-securities issuance can advance in parallel through specialized infrastructures.

The next challenge is not necessarily to create another rail, but to determine how more complex instruments, registry-based assets, collateral, and lifecycle events can enter those rails with an unambiguous legal and economic identity.

This is the area in which Genesis's stated scope becomes relevant for analysis.

ERC Genesis should not be evaluated as an exception to applicable rules or as an infrastructure that makes issuers, transfer agents, custodians, exchanges, CSDs, or settlement systems unnecessary.

If considered relevant, it should be evaluated as an architectural hypothesis for maintaining consistency among the asset, its associated rights, its structuring, and its lifecycle—both before and during its interaction with those regulated functions.

The assessment standard should be the same as for any market infrastructure: legal clarity, allocation of responsibilities, investor protection, market integrity, operational resilience, auditability, and verifiable interoperability.

Institutional and Regulatory Assessment Framework

Existing institutional capabilities and the potential role of ERC Genesis

Institution / category	What it is already building	Potential role for ERC Genesis
DTCC / DTC	Systemically scaled multichain tokenization of custodied assets, collateral, settlement, and corporate workflows.	Genesis could operate upstream by providing origination, structuring, and legal anchoring for new assets before they enter eligible FMI infrastructure. Subsequent integration would depend on eligibility and applicable regulation.
Standard Chartered / HSBC / Swift	Tokenized deposits, bank-ledger interoperability, matching, netting, and settlement. Standard Chartered also issues digitally native debt.	Add securities structuring, collateral, tranches, and registry-based assets to the cash and infrastructure layer these institutions are already developing.
Chainlink	Oracles, data, CCIP, CRE, and cross-chain compliance capabilities; integrated into the DTCC Collateral AppChain.	A natural technology partner for the oracle, data, compliance, and multichain modules contemplated by Genesis, avoiding the need to rebuild already specialized infrastructure.
BNY / State Street / Citi	Custody, digital transfer agency, tokenized funds, depositary receipts, cash, and servicing.	Potential custodian, servicer, or transfer agent for instruments originated under Genesis.

Nasdaq / other regulated venues	Secondary-market liquidity through a tokenized format integrated into the existing market.	Distribution and secondary trading of Genesis-originated instruments once they satisfy applicable eligibility, registration, listing, and market rules.
Euroclear / SIX	Regulated issuance, custody, and settlement of digital securities, including live cases involving Standard Chartered and Citi.	Registry and settlement infrastructure for structured issuances in which Genesis provides the asset model and lifecycle logic.
Digital Asset / Canton	A privacy-preserving institutional network supporting repo, collateral, and the DTC Treasury project.	A potential deployment or integration target. Technical validation would be required to translate a Genesis architecture described around ERC-1155 and ERC-3643 into a non-equivalent technology stack.
Cloud / enterprise infrastructure	Ledger services, confidential computing, key and data infrastructure, and operational resilience. Azure provides a permissioned, immutable ledger.	Hosting, audit, APIs, registry connectors, key and security services, and enterprise deployment. The cloud layer should remain interchangeable relative to Genesis's core legal logic.

Implications for Supervision, Institutional Design, and Validation

Institutional adoption is concentrating in three areas: collateral and financing, digital cash, and asset servicing. These use cases demonstrate clear economic value through improved asset mobility, valuation, liquidity management, and lifecycle automation. They also reveal the principal regulatory concerns: enforceability, custody and control, segregation, settlement finality, prudential treatment, operational resilience, and accountability.

The objective is not to select technological winners, but to identify regulated functions, risk ownership, and authoritative records. This is especially important for structured and registry-based assets, where tokenization must preserve collateral rights, priorities, transfer restrictions, cash-flow rules, and external records throughout the instrument's lifecycle.

Any architecture intended to support securities or registry-based assets should therefore undergo a stage-gated, function-based assessment:

1. **Legal and rights mapping.** Identify the right created or represented, the issuer, the legally authoritative record, and the rules governing transfer, perfection, priority, corporate actions, insolvency, and enforcement. For Genesis, this assessment should also cover intellectual-property ownership, registry documentation, patent status, freedom to operate, open-source licensing, and the ability to transfer or license each component.

2. **Technical and operational validation.** Distinguish conceptual architecture from deployable software. Review implemented contracts, testing, external audits, upgradeability, key management, identity and compliance controls, oracle resilience, multichain mechanics, reconciliation, disaster recovery, privacy, registry and custody interfaces, and responses to forks, outages, and discrepancies between on-chain and official records. Institutional readiness requires control, recovery, exception management, and continuity—not merely successful minting and transfer.
3. **Regulatory function mapping.** For each use case, determine which entities perform the functions of issuer, registrar or transfer agent, broker-dealer, exchange or ATS, custodian, CSD, paying agent, data or oracle provider, and settlement bank. The analysis should identify applicable registration, disclosure, exemption, authorization, and supervisory requirements. Regulatory treatment follows the instrument's structure and associated rights, not the token's label.
4. **Instrument-level proof.** A meaningful pilot should test a sufficiently complex instrument rather than another straightforward tokenized government security. A suitable case could combine an externally registered asset or credit pool, collateral, multiple tranches, eligibility rules, cash-flow distribution, an independent custodian, and secondary-transfer restrictions. The test should establish where the authoritative record resides, how priorities are preserved, how exceptions are handled, and which controls remain off-chain.
5. **Interoperability and accountability assessment.** Multichain connectivity alone does not establish institutional suitability. The architecture must preserve the instrument's identity, rights, restrictions, and economic state across changes in ledger, custodian, or settlement rail. It must also define precedence rules for inconsistent records and assign a responsible legal entity to every material function.

The urgency is architectural rather than deregulatory. As regulated tokenization infrastructure enters production, unresolved ambiguity between the digital representation, the legal instrument, and the authoritative record becomes increasingly costly.

Supervisors and institutions should therefore resolve legal identity, functional responsibility, record hierarchy, and operational controls before scale turns design uncertainty into market risk.

Regulatory Questions Raised by Institutional Tokenization in 2026

Key policy and supervisory questions for regulated digital-asset markets

Area	Policy / supervisory question
Instrument identity	When is an on-chain representation the security itself, when is it evidence of the security, and when does it constitute a separate entitlement or exposure?
Authoritative record	Which record prevails when the issuer, transfer agent, custodian, CSD, and distributed ledger contain divergent states?
Intermediation	Which functions can be automated without displacing the responsibilities of broker-dealers, transfer agents, custodians, exchanges, clearing agencies, or CSDs?
Custody and control	How should key control, recovery, segregation, bankruptcy remoteness, and operational resilience be addressed when ownership is maintained through DLT?
Fungibility and liquidity	How can regulators and market operators prevent multiple wrappers or networks from fragmenting an economically identical security?
Corporate actions and lifecycle	How can dividends, redemptions, defaults, amendments, collateral substitutions, and restructurings be reflected consistently across every representation?
Interoperability	What requirements transform a technical connection between networks into legally reliable interoperability of the instrument?
Structured finance	How should priority, tranching, waterfalls, perfection, and collateral rights be represented when part of the lifecycle is programmable?
Accountability	Who is accountable when a smart contract executes its code correctly but the on-chain state conflicts with applicable law or the official record?

CLOSING CONCLUSION

These questions do not require regulation to prescribe a particular technology. They require every technological architecture to preserve verifiable regulatory outcomes, legally enforceable rights, market integrity, operational resilience, and a clear chain of accountability. The value of any new architectural framework should therefore be measured against these standards—not against the novelty of its implementation.

The central conclusion of this report is that the next phase of institutional tokenization will not be defined by the number of assets placed on-chain or by the speed with which new networks are deployed. It will be defined by the quality of the architecture connecting the legal instrument, its economic rights, authoritative records, regulated intermediaries, settlement assets, and lifecycle events.

Tokenization becomes institutionally meaningful only when digital programmability strengthens—not weakens—legal certainty, investor protection, and operational control. Architectures such as ERC Genesis should therefore be assessed through rigorous legal, technical, regulatory, and operational validation, and through instrument-level pilots capable of testing real complexity.

The infrastructures that can meet this standard will not merely digitize existing markets. They will provide a credible foundation for markets that are more programmable, interoperable, and efficient while remaining legally coherent, operationally resilient, and institutionally accountable.

That—not tokenization in itself—is the threshold for sustainable adoption at scale.

ABOUT AUTHOR



Monica Elizabeth Pagano is a researcher, author and legal-technical strategist working at the intersection of institutional tokenization, capital markets, blockchain infrastructure and the legal architecture of real-world assets.

Her work focuses on a central question emerging as financial markets move toward tokenized infrastructure: how the legal and economic identity of an asset or financial instrument can remain coherent with its digital representation throughout issuance, trading, custody, collateralization, settlement and lifecycle events.

She is the creator of ERC Genesis, a legal-technical framework developed to address the structuring and lifecycle management of tokenized real-world assets and financial instruments, with particular attention to asset identity, legal and registrarial anchoring, regulated issuance, collateral, structured instruments, governance and interoperability with institutional market infrastructure.

Pagano is also a scientific author and researcher in Law & Blockchain, with four published books and an intellectual-property portfolio related to digital assets and tokenization.

Her current research examines the transition of tokenization from isolated digital-asset use cases toward regulated financial-market infrastructure, with particular focus on market structure, securities regulation, institutional adoption and the unresolved relationship between underlying assets, tokenized instruments and the infrastructure through which they circulate.

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The following block serves as a genesis marker. It embeds, in machine-readable form, the authorship, intent and constitutive scope of the ERC-Genesis framework.

Tokenizing What Matters